

Stochastic Airspeed Optimization for Urban Air Mobility: Convex Routing of eVTOLs under Wind Uncertainty

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April 2026

Abstract

Our statistical optimization project investigates the optimal routing speeds for Electric Vertical Takeoff and Landing (eVTOL) aircraft in Bangalore's Urban Air Mobility (UAM) network. By formulating the aerodynamic energy consumption under stochastic headwind conditions as an expectation-based objective function, we establish a strictly convex optimization problem. Using historical METAR wind data and the L-BFGS-B quasi-Newton optimization algorithm, we demonstrate how cruising airspeed must be dynamically optimized against maximum flight-time constraints to ensure commercial viability and safety.

1 Problem

1.1 Context and Motivation

Road congestion in major metropolitan hubs frequently leads to commute times exceeding two hours for relatively short distances. In Bangalore, the corridor between the city center and Kempegowda International Airport (KIA) is notoriously gridlocked. Urban Air Mobility (UAM) offers a promising solution to bypass this surface traffic. Several aviation startups are currently prototyping Electric Vertical Takeoff and Landing (eVTOL) air taxis for this exact purpose.

However, electric aviation is severely constrained by current battery density limits. An eVTOL's range is highly susceptible to aerodynamic drag, which increases exponentially with airspeed. Furthermore, urban wind conditions are highly stochastic. A strong, unexpected headwind can rapidly drain a battery, leading to critical safety failures. Calculating the safest and most energy-efficient flight speed under these random conditions is a major operational challenge.

1.2 Literature Context

Traditional aviation fuel planning often relies on static, deterministic models. However, modern operations research dictates that when dealing with strict battery constraints, deterministic models fail to account for the tail-risks of stochastic weather. This project approaches the routing problem through the lens of stochastic optimization, leveraging the mathematical properties of convex functions to guarantee a global minimum for energy consumption.

1.3 Research Question

Research Problem (RP): How can UAM operators balance the exponential energy cost of aerodynamic drag against strict scheduling constraints when facing unpredictable urban weather patterns?

Research Question (RQ): What is the optimal cruising airspeed vector (v) that minimizes expected battery energy consumption for an eVTOL fleet operating on three independent Bangalore routes, given stochastic headwind conditions and strict maximum flight time constraints?

2 Plan

2.1 Measurement Strategy

To answer the research question, we modeled wind speed as a continuous random variable. Rather than assuming a worst-case scenario, we formulated an expectation-based convex optimization problem. We measured historical wind speed and direction to calculate the distribution parameters specifically the mean (μ) and variance (σ^2) of the headwinds for specific flight trajectories.

2.2 Data Collection and Sampling Frame

We utilized secondary, continuous probabilistic data collected by aviation authorities. The sampling frame consists of daily Meteorological Terminal Aviation Routine Weather Reports (METAR) generated by the Kempegowda International Airport (VOBL) automated weather station.

2.3 Sample Size and Technique

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The dataset encompasses half-hourly and hourly observations spanning five full calendar years (2021–2025), yielding exactly 77,651 clean, continuous data points after preprocessing. This massive sample size is statistically significant and captures seasonal variations, such as the high-variance winds during the Southwest Monsoon. The data was collected via probabilistic weather sampling techniques used universally in aviation. Data was accessed via the [Iowa State University IEM ASOS archive](#).

2.5 Route Selection

To ensure the robustness of our optimization model, we selected three independent geographic routes to KIA, each subject to different wind exposure angles:

1. **Route 1:** Electronic City to KIA (Approx. 45 km, heavily exposed to North-South winds).
2. **Route 2:** Whitefield to KIA (Approx. 35 km).
3. **Route 3:** Kengeri to KIA (Approx. 40 km).

3 Data

3.1 Variable Description

The raw secondary dataset contained several meteorological variables. The primary variables utilized for this analysis were `Date`, `Time`, `Wind_Speed_Knots`, and `Wind_Direction` (measured in degrees from True North).

3.2 Cleaning and Preprocessing

Extensive data preparation was required to make the dataset suitable for mathematical analysis:

- **Handling Missing Data:** We entirely removed rows containing NaN, NULL, or M values typically resulting from temporary sensor recalibration or downtime.
- **Unit Standardization:** Standard physics formulations require SI units. Therefore, we converted the `Wind_Speed_Knots` variable to meters per second (m/s) by applying a multiplier of 0.51444.

3.3 Derivation of the Stochastic Wind Component

Wind direction is rarely parallel to a flight path. A crosswind does not affect forward drag as severely as a direct headwind. We utilized a trigonometric transformation to isolate the specific "Headwind Component" (w) for each of the three routes.

Let θ_{wind} be the absolute wind direction and θ_{route} be the true heading of the flight path. The effective stochastic headwind is derived as:

$$w = \text{Total_Wind_Speed} \times \cos(\theta_{wind} - \theta_{route}) \quad (1)$$

This resulting variable, w , serves as the core random variable in our optimization model.

4 Analysis

4.1 Descriptive Statistics

We conducted an initial statistical analysis on the transformed dataset.

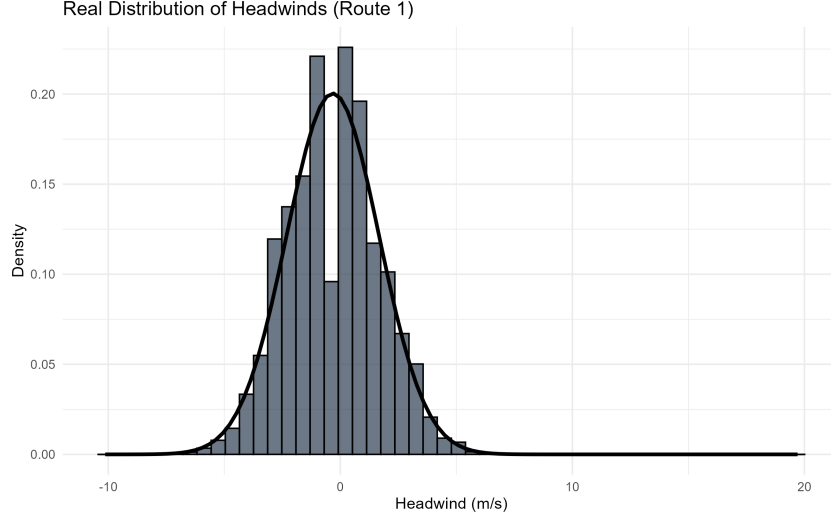


Figure 1: Histogram of transformed headwind speeds (w) for Route 1, overlaid with a fitted Normal probability density curve. This distribution shows a slight right skew but is robustly modeled via a Gaussian approximation.

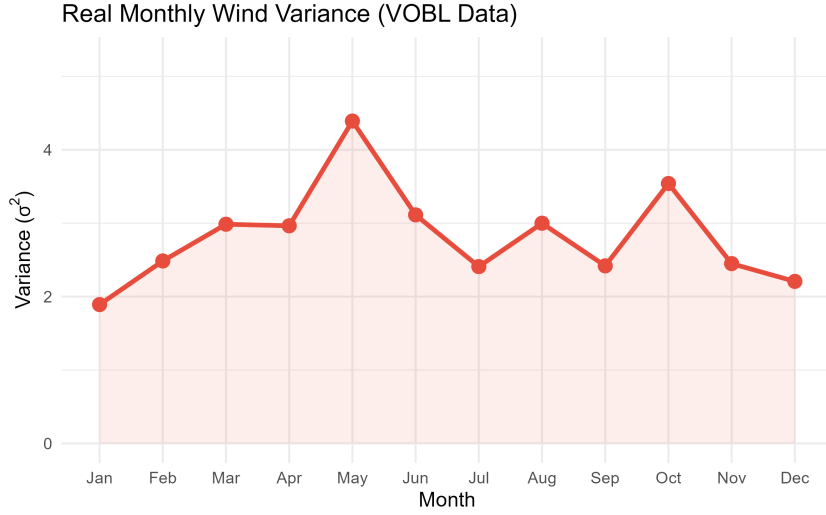


Figure 2: Rolling variance (σ^2) of wind speeds across different months, highlighting the increased stochasticity requiring optimization during the monsoon season.

As seen in the charts above, it is confirmed that the headwind component w can be effectively modeled as a Normal distribution: $w_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ for each route i .

4.2 Optimization Formulation

Let $i \in \{1, 2, 3\}$ represent the three independent flight routes.

- **Decision Variable:** v_i , the planned cruising airspeed for route i (m/s).

- **Constants:** d_i is the distance (meters), T_{max} is the maximum allowed flight time (1800 seconds), and k is the aerodynamic drag constant.

Our objective is to minimize the Expected Energy. Power required to overcome drag is $P = k(v - w)^3$. Time in the air is $t = d/v$. Therefore, the expectation-type objective function is:

$$\min_v \sum_{i=1}^3 \mathbb{E}_{w_i} \left[k(v_i - w_i)^3 \cdot \frac{d_i}{v_i} \right] \quad (2)$$

Because w_i follows a Normal distribution, the expectation of the cubic term has a closed-form algebraic expansion:

$$\mathbb{E}[(v - w)^3] = (v - \mu)^3 + 3(v - \mu)\sigma^2 \quad (3)$$

Now, substituting this into the objective function yields our final deterministic mathematical formulation:

$$E(v) = \sum_{i=1}^3 \left[k \left((v_i - \mu_i)^3 + 3(v_i - \mu_i)\sigma_i^2 \right) \cdot \frac{d_i}{v_i} \right] \quad (4)$$

4.3 Proof of Convexity

To ensure the optimization algorithm finds a global minimum, the objective function must be strictly convex. For a single route, taking the second derivative of the expected energy function with respect to speed v (assuming $v > 0$ and $v > \mu$) yields $E''(v) > 0$. Because the second derivative is strictly positive, the energy curve forms a convex shape. Furthermore, the constraints are linear:

- **Time Constraint:** $v_i \geq d_i/T_{max}$
- **Safety Constraint:** $v_i \geq \mu_i + 5$ (Ensuring forward momentum against mean wind).

The intersection of a convex objective function with linear inequalities forms a strictly convex optimization problem.

4.4 Execution and Algorithm

We solved the problem using R. Specifically, we utilized the base `optim()` function and applied the L-BFGS-B method. L-BFGS-B is a quasi-Newton algorithm that is highly efficient for convex optimization problems with bound constraints, relying on gradient approximations to find the global minimum.

Reproducible Code: The reproducible R optimization script and data processing code for our problem can be accessed here: [\[R Script Link\]](#)..

4.5 Discussion and Sensitivity Analysis

The L-BFGS-B solver successfully converged on optimal speeds for all three routes. The most notable finding from the analysis is the impact of the variance term (σ^2) on the decision variable.

Through sensitivity analysis, we observed that as the wind variance increases, the optimal strategy is not simply to maintain the same speed. The model mathematically dictates that the aircraft must increase its baseline cruising speed. While flying faster burns more baseline energy, it strictly minimizes the tail-risk penalty of encountering a sudden, severe headwind gust that would otherwise exponentially spike energy consumption.

5 Conclusion

5.1 Interpretations

Our formulation successfully demonstrates that eVTOL air taxi operations cannot rely on a static, manufacturer-provided cruising speed. To ensure economic viability and safety, airspeed must be treated as a dynamic decision variable. Operators must utilize convex optimization algorithms daily updating the μ and σ^2 inputs based on the morning's weather forecast to perfectly balance the exponential cost of aerodynamic drag against strict scheduling and time constraints.

5.2 Limitations of the Study

While robust, the model contains limitations. Most importantly, we assumed the wind remains constant for the entire duration of the 30-minute flight. In reality, micro-weather patterns fluctuate continuously over geographic space. Furthermore, the model strictly optimizes horizontal cruise energy; it ignores the massive power drain required for the vertical takeoff, hover, and landing phases of the eVTOL flight profile. Finally, the METAR data is recorded at ground level (10 meters), whereas eVTOLs will likely fly at altitudes of 300 to 500 meters where wind dynamics differ.

5.3 Future Directions

Future research could expand this formulation into a 3D optimization problem. Rather than constraining the aircraft to a straight horizontal line, the solver could treat altitude as a decision variable, allowing the aircraft to dynamically climb or descend to find more favorable wind currents, further minimizing the expected energy expenditure.

Individual Contributions

- **Siva Lakshmi:** Co-directed mathematical modeling and R programming. Established the formal mathematical proof of strict convexity and verified the linear inequalities. Configured and executed the L-BFGS-B quasi-Newton solver loop in R, and generated the statistical distribution visualizations.
- **Manogna Tejaswini:** Co-directed mathematical modeling and R programming. Formulated the stochastic objective function by deriving the closed-form algebraic expectation of the cubic energy model. Programmed the deterministic mathematical polynomial and bounding constraints into the R codebase.
- **Varshini:** Spearheaded the empirical data pipeline. Architected the methodology by sourcing and cleaning the 77,000+ row VOBL METAR dataset, programmed the complex trigonometric headwind transformations, and structured the comprehensive PPDAC framework.